Individual Reflection

Medical Question Answering on PubMedQA
EMATM0067 – Introduction to AI and Text Analytics
Group 16

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1 Contributions and Activity

I led the integration and advanced-modeling for Group 16. I developed all the advanced transformer experiments and the core minority-class intervention suite in a stepwise eight-week period. This included the PubMedBERT fine-tuning experiment (Q+Ctx+Ans) (A1g), LoRA adaptation (A2i), zero-shot BART-MNLI (A2j), class-weighted focal loss (M2), validation-tuned *maybe*-thresholding (M1/M3), ordinal-aware loss (M4), and the two-stage hierarchical classifier (M5).

I wrote the results exports, tables and figures. I cleaned the master notebook for reproducibility, maintained the GitHub repository and added Overleaf report code. My contributions are listed in Figure 1. I made 70 commits with 100,367 additions and 26,690 deletions. I was ranked first contributor and active throughout the project.

I helped to tie together the experimental code, the final report narrative, and to make the results reproducible and consistent.

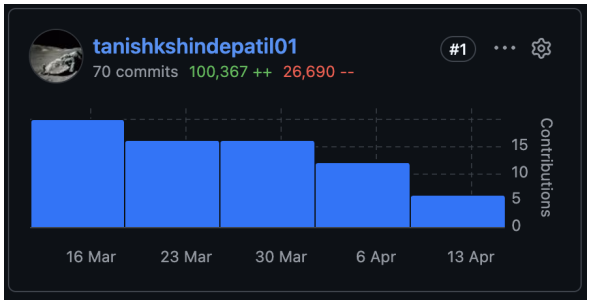

Figure 1: GitHub contribution evidence for `tanishkshindepatil01`, showing 70 commits and the highest contributor rank.

2 Technical Understanding

The most technical challenge I was closely involved with was minority-class recovery in PubMedQA. Since the *maybe* label only represented 11% of the dataset (110/1,000), looking at overall accuracy hides misclassifications on uncertainty. To remedy this, I also investigated models and training objectives that would improve recovery of *maybe*-class while tracking the trade-off against the overall macro-F1 objective, broken out in Figure 2.

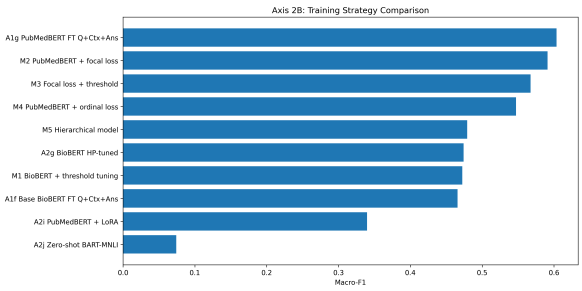

Figure 2: Macro-F1 for the advanced transformer and *maybe*-class experiments I implemented.

First, we fine-tuned PubMedBERT on Q + Ctx + Ans (A1g), which yielded our best-performing model with test macro-F1 = 0.604 and test accuracy = 75.4%. It considerably outperformed TF-IDF + LR per the McNemar's test with $b = 164$, $c = 40$, $p < 0.001$. However, the confusion matrix indicated only 9/55 true *maybe* cases were recovered, and 46 were forced into *yes*/*no*. This taught me that a high headline score is not enough for biomedical QA, because uncertainty suppression can still be clinically risky, see Figure 3.

Using class-weighted focal loss (M2), the *maybe*-F1 was the best, increasing from 0.192 to 0.237, and the macro-F1 decreased from 0.604 to 0.591, Figure 4. I applied focal loss to down-weight easy majority examples. I tested the *maybe* softmax thresholds on M1/M3, the ordinal-aware loss (M4), and the hierarchical *maybe*-vs-definitive classifier (M5). These did not generalize well: M3's *maybe*-F1 fell to 0.094 and M5 failed to learn a useful uncertainty detector.

My main technical contribution is that, for PubMedQA, the current bottleneck is not model capacity but evidence-sufficiency reasoning in the pres-

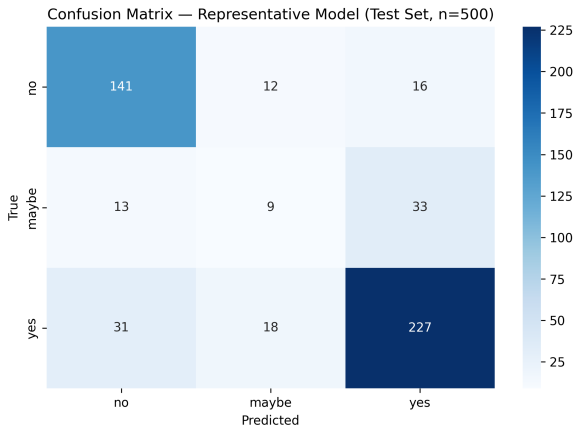

Figure 3: A1g confusion matrix showing that 46/55 true *maybe* cases were forced into *yes*/*no*.

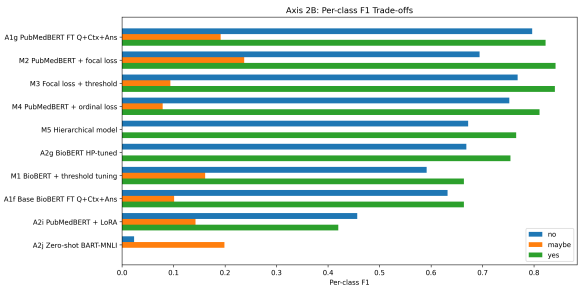

Figure 4: Per-class F1 trade-off: A1g maximises macro-F1, while M2 gives the best *maybe*-F1.

ence of extreme minority-class supervision. A1g is most effective, but M2 achieves better performance when missing uncertainty is more costly.

3 Teamwork

We coordinated through a Jira board, a shared GitHub repository, weekly team meetings, and a group chat. I coordinated the experiment names, notebook outputs, exported result files, figures, and report tables so that other people could integrate their code with mine, and so it would be clear who was doing what.

All authors were involved in the paper. Abdullah: data loading, preprocessing, label encoding, constructing text representations, preparing splits and CV objects, sanity checks. Peidong: our baseline, TF-IDF + LR, our main model, BioBERT embeddings + LR, BioBERT fine-tuning. Zhengjie implemented classical Axis-2 models, tuned LR + calibrated SVM + Multinomial NB + SMOTE + LR + soft voting, while Abhigyan helped with error analysis, confusion-matrix, interesting misclassified examples, and checking results. I built the advanced transformer/*maybe*-class models and the final integration.

I also tried to create a culture of inclusion in which people owned their own stage from beginning to end. When the stages were joined, I checked that tables, figures and model labels were in accordance with the notebook and report. This reduced the number of integration errors near submission.